

# Elementary Problems

## Exponent

## Solutions

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## Order

1. We check that

$$3^k \equiv 3, 2, 6, 4, 5, 1 \pmod{7}$$

for  $k = 1, 2, 3, 4, 5, 6$ . Therefore, the order of 3 modulo 7 is 6.

Similarly, we check that

$$3^k \equiv 3, 9, 1 \pmod{13}$$

for  $k = 1, 2, 3$ . Therefore, the order of 3 modulo 13 is 3.

2. Let  $d$  be the order of 5 modulo 64. By Euler-Fermat theorem, we have  $d \mid \varphi(64) = 32$ . Thus,  $d$  must be a power of 2. Note that

$$\begin{aligned} 5^4 &= 25^2 = 625 \equiv 49 \pmod{64}, \\ 5^8 &\equiv 49^2 \equiv (-15)^2 = 225 \equiv 33 \pmod{64}, \\ 5^{16} &\equiv 33^2 = 1089 \equiv 1 \pmod{64}. \end{aligned}$$

This shows  $d = 16$ .

3. (a) Since  $2^1 \equiv 2 \pmod{7}$ ,  $2^2 \equiv 4 \pmod{7}$  and  $2^3 \equiv 1 \pmod{7}$ , the order of 2 modulo 7 is 3. Thus, only those  $n$ 's which are multiples of 3 satisfy  $7 \mid 2^n - 1$ .
- (b) From the proof of (a),  $2^n + 1$  can only be congruent to 3, 5, 2 modulo 7. Thus, it cannot be a multiple of 7.
4. Clearly,  $n > 1$  and  $(a, n) = 1$ . Let  $d$  be the order of  $a$  modulo  $n$ . Then  $d \mid n - 1$  but  $d \nmid \frac{n-1}{p}$  for any prime divisor  $p$  of  $n - 1$ . Thus, the only possibility is  $d = n - 1$ . Since  $d \mid \varphi(n)$ , we have  $n - 1 \leq \varphi(n)$ . This yields  $\varphi(n) = n - 1$  as  $\varphi(n) < n$ . In that case,  $n$  must be a prime.
5. Note that the decimal representation of any fraction  $\frac{a}{b}$  has the same period as  $\frac{a}{2^s 5^t b}$  for any nonnegative integers  $s$  and  $t$ . Therefore, it suffices to consider the case  $(n, 10) = 1$ . Also, we may assume  $0 \leq x < 1$  by removing the integral part.
- Let  $d$  be the order of 10 modulo  $n$ . Then  $d \leq n - 1$ , and there exists  $k \in \mathbb{N}$  such that  $10^d - 1 = kn$ . This gives

$$x = \frac{m}{n} = \frac{km}{10^d - 1}.$$

This shows  $x$  has period at most  $d$ . Indeed, if we let  $km = \overline{a_d a_{d-1} \cdots a_1}$ , then

$$x = \dot{a}_d a_{d-1} \cdots a_2 \dot{a}_1.$$

6. Clearly,  $p$  is odd. Let  $d$  be the order of 2 modulo  $p$ . Then  $d \mid q$ . This yields  $d = 1, q$ . The case  $d = 1$  is rejected since  $2 \not\equiv 1 \pmod{p}$ . Hence,  $d = q$ .

By Fermat little theorem, we have  $d \mid p - 1$ . This shows  $p \equiv 1 \pmod{q}$ .

7. Obviously,  $(m, n) = (2, 1)$  is a solution. We claim that there is no solution for  $n \geq 2$ . By considering modulo 9, we have

$$2^m \equiv 1 \pmod{9}.$$

It follows from the pattern  $2^m \equiv 2, 4, 8, 7, 5, 1 \pmod{9}$  that  $6 \mid m$ .

Let  $m = 6k$ . Then the equation becomes  $2^{6k} - 3^n = 1$ . By considering modulo 7, since  $2^6 = 64 \equiv 1 \pmod{7}$ , we have

$$1 - 3^n \equiv 1 \pmod{7}.$$

This yields  $7 \mid 3^n$ , which is impossible.

Therefore, the only solution is  $(m, n) = (2, 1)$ .

8. Obviously,  $(m, n) = (1, 1), (2, 3)$  are solutions, while  $n = 2$  yields no solution. We claim that there is no solution for  $n \geq 4$ . By considering modulo 16, we have

$$3^m \equiv 1 \pmod{16}.$$

It follows from the pattern  $3^m \equiv 3, 9, 11, 1 \pmod{16}$  that  $4 \mid m$ .

Let  $m = 4k$ . Then the equation becomes  $3^{4k} - 2^n = 1$ . By considering modulo 5, since  $3^4 = 81 \equiv 1 \pmod{5}$ , we have

$$1 - 2^n \equiv 1 \pmod{5}.$$

This yields  $5 \mid 2^n$ , which is impossible.

Therefore, the only solutions are  $(m, n) = (1, 1), (2, 3)$ .

9. Note that  $n = 1$  yields no solution. For  $n \geq 2$ . By considering modulo 9, we have

$$5^m \equiv 1 \pmod{9}.$$

It follows from the pattern  $5^m \equiv 5, 7, 8, 4, 2, 1 \pmod{9}$  that  $6 \mid m$ .

Let  $m = 6k$ . Then the equation becomes  $5^{6k} - 12^n = 1$ . By considering modulo 7, since  $5^6 \equiv 1 \pmod{7}$  by Fermat little theorem, we have

$$1 - 5^n \equiv 1 \pmod{7}.$$

This yields  $7 \mid 5^n$ , which is impossible.

Therefore, there is no solution.

10. By considering modulo 8, we have  $3^m - 1 \equiv 0 \pmod{8}$ . This shows  $m$  is even. Let  $m = 2k$ . Then

$$17^n = 3^{2k} - 64 = (3^k - 8)(3^k + 8).$$

The numbers  $3^k - 8$  and  $3^k + 8$  are positive integers. Since their product is a power of 17, each of them is a power of 17. As

$$(3^k - 8, 3^k + 8) = (3^k - 8, 16),$$

only one of them can be a multiple of 17. Hence, the only possible case is  $3^k - 8 = 1$ , which gives  $k = 2$  and  $n = 1$ . Then  $(m, n) = (4, 1)$ .

Checking:  $3^4 - 17^1 = 64$ .

11. Obviously,  $(m, n) = (1, 2)$  is a solution. We claim that there is no solution for  $m \geq 2$ . By considering modulo 9, we have

$$5^n \equiv 4 \pmod{9}.$$

It follows from the pattern  $5^n \equiv 5, 7, 8, 4, 2, 1 \pmod{9}$  that  $n \equiv 4 \pmod{6}$ .

Using  $n \equiv 4 \pmod{6}$  and Fermat little theorem, we have

$$5^n \equiv 5^4 = 25^2 \equiv 4^2 \equiv 2 \pmod{7}.$$

This yields  $3^m \equiv 1 \pmod{7}$  and hence  $6 \mid m$ .

Next, by considering modulo 8, as  $m$  and  $n$  are even from above, we get

$$1 + 22 \equiv 1 \pmod{8}.$$

This is a contradiction.

Therefore, the only solution is  $(m, n) = (1, 2)$ .

12. By considering modulo 3, we have  $1 \equiv 2^z \pmod{3}$ . This shows  $z$  is even. Let  $z = 2k$ . Then

$$3^x = 5^z - 4^y = 5^{2k} - 2^{2y} = (5^k - 2^y)(5^k + 2^y).$$

The numbers  $5^k - 2^y$  and  $5^k + 2^y$  are positive integers. Since their product is a power of 3, each of them is a power of 3. As

$$(5^k - 2^y, 5^k + 2^y) = (5^k - 2^y, 2 \cdot 5^k),$$

only one of them can be a multiple of 3. Hence, the only possible case is  $5^k - 2^y = 1$ .

If  $y \geq 3$ , we have  $5^k \equiv 1 \pmod{8}$  so that  $k$  is even. Let  $k = 2\ell$ . Then  $5^{2\ell} - 2^y = 1$ . As

$$5^{2\ell} = 25^\ell \equiv 1 \pmod{3},$$

we have  $1 - 2^y \equiv 1 \pmod{3}$ . This implies  $2^y$  is divisible by 3, which is impossible.

By checking  $y = 1, 2$ , we find that the only solution is  $(x, y, z) = (2, 2, 2)$ .

13. Consider modulo 17. One can check that

$$2^x \equiv 2, 4, 8, 16, 15, 13, 9, 1 \pmod{17} \quad (1)$$

when  $x = 1, 2, \dots$ , and

$$y^4 \equiv 0, 1, 4, 13, 16 \pmod{17}. \quad (2)$$

Comparing (1) and (2), since  $2^x \equiv y^4 \pmod{17}$ , the only possibilities of the remainders are 1, 4, 13, 16. From (1), this only happens when  $x \equiv 0, 2, 4, 6 \pmod{8}$ . This shows  $x$  must be even.

Let  $x = 2n$ . Then the equation becomes  $y^4 - 2^{2n} = 17$ . This is the same as

$$(y^2 - 2^n)(y^2 + 2^n) = 17.$$

The only possibility is  $y^2 - 2^n = 1$  and  $y^2 + 2^n = 17$ . This gives  $y^2 = \frac{1+17}{2} = 9$  so that  $y = 3$ . Correspondingly, we have  $x = 6$ .

Checking:  $2^6 + 17 = 81 = 3^4$ .

14. Firstly, suppose  $x \geq 6$ . Then we have  $3^y + 5 = 2^x \equiv 0 \pmod{64}$ . The powers of 3 modulo 64 has the following pattern

$$3^k \equiv 3, 9, 27, 17, 51, 25, 11, 33, 35, 41, 59, 49, 19, 57, 43, 1 \pmod{64}.$$

This shows the order of 3 modulo 64 is 16. Also, from  $3^y \equiv 59 \pmod{64}$ , we have  $y \equiv 11 \pmod{16}$ .

By Fermat little theorem, we have  $3^{16} \equiv 1 \pmod{17}$ . Hence,

$$2^x = 3^y + 5 \equiv 3^{11} + 5 \equiv 27^3 \cdot 9 + 5 \equiv 10^3 \cdot 9 + 5 = 9005 \equiv 12 \pmod{17}.$$

However, we find that

$$2^k \equiv 2, 4, 8, 16, 15, 13, 9, 1 \pmod{17}.$$

There is no solution for  $x \geq 6$ .

Checking the small cases  $x = 1, 2, 3, 4, 5$ , we find the only solutions  $(x, y) = (3, 1), (5, 3)$ .

15. **Solution 1:**

By Dirichlet's theorem, we can choose distinct primes  $p$  and  $q$  with  $p, q \equiv -1 \pmod{n}$ . The number  $r = p^2 q^2$  satisfies  $r \equiv 1 \pmod{n}$ . Also, it can be decomposed as  $(p^2)(q^2)$  and  $(pq)(pq)$ , where each of  $p^2, q^2, pq$  is congruent to 1 modulo  $n$ . These numbers are indecomposable in  $V_n$  as  $p, q \notin V_n$ .

**Solution 2:**

We first observe that there are infinitely many primes not of the form  $kn + 1$ . Indeed, if there is a finite number of them, then we let  $P$  be their product and consider  $Pn - 1$ . It

must contain a prime divisor not of the form  $kn + 1$  (otherwise it should be congruent to 1 modulo  $n$ ) which cannot be a divisor of  $P$ . This is a contradiction.

By the pigeonhole principle, we can choose two primes  $p \equiv q \not\equiv 1 \pmod{n}$ . Let  $d$  be the order of  $p$  (and hence  $q$ ) modulo  $n$ . Consider  $r = p^d q^d$ . Note that  $r \equiv 1 \pmod{n}$  as  $p^d \equiv q^d \equiv 1 \pmod{n}$ . It can be decomposed as  $(p^{d-1}q)(pq^{d-1})$  and  $(p^d)(q^d)$ . Each of the numbers  $p^{d-1}q, pq^{d-1}, p^d, q^d$  is congruent to 1 modulo  $n$ . Also, they are indecomposable in  $V_n$  as  $p^i q^j \equiv p^{i+j} \not\equiv 1 \pmod{n}$  for  $1 \leq i + j \leq d - 1$ .

16. If both  $m$  and  $n$  are even, then  $4 \mid mn$  so that  $4 \mid 3^m + 1$ . However, we have

$$3^m + 1 \equiv 1 + 1 \equiv 2 \pmod{4},$$

which is a contradiction.

WLOG assume  $m$  is odd. For  $m = 1$ , we need  $n \mid 3^1 + 1 = 4$ . The only possibilities are  $n = 1, 2, 4$ . One checks that only  $n = 1, 2$  fulfill the conditions.

For  $m \geq 3$ , let  $p$  be the smallest prime divisor of  $m$ . As  $p \mid 3^m + 1$ , we have  $(p, 3) = 1$ . Let  $d$  be the order of 3 modulo  $p$ . From

$$3^{2m} = (3^m)^2 \equiv (-1)^2 = 1 \pmod{p},$$

we have  $d \mid 2m$ . By Fermat little theorem, we also have  $d \mid p - 1$ . Note that

$$(2m, p - 1) = (2, p - 1) = 2.$$

Indeed, if  $m$  and  $p - 1$  have a common prime divisor  $q$ , then  $q$  is a divisor of  $m$  smaller than  $p$ . This contradicts the minimality of  $p$ . From this, we find that  $d \mid 2$ , i.e.  $d = 1, 2$ . By definition,

$$3^1 \equiv 1 \pmod{p} \quad \text{or} \quad 3^2 \equiv 1 \pmod{p}.$$

Both are impossible since  $p$  is an odd prime.

Therefore, the only solutions are  $(m, n) = (1, 1), (1, 2), (2, 1)$ .

### Lifting the Exponent Lemma

17. By simple checking, we find that the order of 2 modulo 5 is 4. Also, we have  $5 \parallel 2^4 - 1$ . By the lifting the exponent lemma, we have

$$5^{k+1} \parallel 2^{4 \cdot 5^k} - 1$$

for any integer  $k \geq 0$ .

Let  $d$  be the order of 2 modulo  $5^8$ . By Euler-Fermat theorem, we have  $d \mid \varphi(5^8) = 4 \cdot 5^7$ . As  $2^d \equiv 1 \pmod{5}$ , we must have  $4 \mid d$ . Thus,  $d$  can only be one of

$$4, 4 \cdot 5, 4 \cdot 5^2, \dots, 4 \cdot 5^7.$$

From above, we know that  $5^8 \nmid 2^{4 \cdot 5^6} - 1$ . Therefore,  $d$  does not divide  $4 \cdot 5^6$ . It follows that the only possibility is  $d = 4 \cdot 5^7$ .

18. It means we need  $1978^m \equiv 1978^n \pmod{1000}$ . We first consider modulo 125. This is the same as  $1978^{n-m} \equiv 1 \pmod{125}$ . Thus,  $n-m$  is at least the order  $d$  of 1978 modulo 125.

Since  $1978 \equiv 3 \pmod{5}$ , it is easy to see that the order of 1978 modulo 5 is 4. Thus  $4 \mid d$ . We check that  $1978^4 \equiv 3^4 \not\equiv 1 \pmod{25}$ . Thus,  $5 \parallel 1978^4 - 1^4$ . By the lifting the exponent lemma, we obtain  $5^2 \parallel 1978^{4 \cdot 5} - 1^{4 \cdot 5}$  and  $5^3 \parallel 1978^{4 \cdot 25} - 1^{4 \cdot 25}$ . It follows that  $d$  divides 100 but  $d$  does not divide 20. Hence, we must have  $d = 100$ .

It suffices to consider the case when  $n \geq 3$ , since otherwise we cannot have  $d \mid n-m$ . In that case,  $1978^m \equiv 1978^n \pmod{8}$  if and only if  $m \geq 3$  as  $2 \parallel 1978$ . Hence, we have  $m+n \geq m+m+100 \geq 106$ . It attains the smallest value when  $m = 3$  and  $n = m+100 = 103$ .

19. We have

$$n = \frac{10^{3^{100}} - 1}{3}.$$

Therefore,  $v_3(n) = v_3(10^{3^{100}} - 1) - 1$ . By the lifting the exponent lemma, we get

$$v_3(10^{3^{100}} - 1) = v_3(10 - 1) + v_3(3^{100}) = 2 + 100 = 102.$$

Hence,  $v_3(n) = 101$ . This shows the highest power of 3 that divides  $n$  is  $3^{101}$ .

20. (1) All even integer  $n$ . Note that  $3 \equiv 3 \pmod{4}$  and  $3^2 \equiv 1 \pmod{4}$ . Therefore, the order of 3 modulo 4 is 2. Hence, for  $a = 3$ , only even  $n$  satisfies  $4 \mid 3^n - 1$ . Conversely, for any even  $n$ , we have  $4 \mid 1^n - 1$  and  $4 \mid 3^n - 1$ . Thus,  $n$  can be any even number.

- (2) All multiples of  $2^{2015}$ . For  $a > 1$ , suppose  $2^\alpha \parallel a^2 - 1$ . For even  $n$ , we have

$$v_2(a^n - 1) = v_2(a^2 - 1) + v_2(n) - 1 = \alpha + v_2(n) - 1$$

by the lifting the exponent lemma.

We claim that the condition is satisfied when  $2^{2015} \mid n$ . Indeed, since  $a^2 \equiv 1 \pmod{8}$  for any odd  $a$ , we have  $\alpha \geq 3$ . Therefore,  $v_2(a^n - 1) \geq 3 + 2015 - 1 = 2017$  if  $2^{2015} \mid n$ . This shows  $2^{2017} \mid a^n - 1$ .

Conversely, if we consider  $a = 3$ , then  $\alpha = 3$  so that

$$v_2(3^n - 1) = v_2(n) + 2.$$

Thus,  $2^{2017} \mid 3^n - 1$  if and only if  $v_2(n) \geq 2015$ , i.e.  $2^{2015} \mid n$ . Thus,  $n$  can be any multiple of  $2^{2015}$ .

21. Since  $3 \mid 5 + 1$ , we have

$$v_3(5^n + 1) = v_3(5 + 1) + v_3(n) = 1 + v_3(n).$$

If  $\frac{5^n + 1}{3^n}$  is an integer, then  $v_3(5^n + 1) \geq n$ . This yields  $v_3(n) + 1 \geq n$ . This means  $n$  is divisible by  $3^{n-1}$ , and thus

$$3^{n-1} \leq n.$$

This holds for  $n = 1$ , and we can check that  $\frac{5+1}{3} = 2$  is an integer. For  $n \geq 2$ , we have

$$3^{n-1} = (1+2)^{n-1} \geq 1 + \binom{n-1}{1} 2 = 2n - 1 > n$$

by binomial expansion. This is a contradiction.

Therefore, the only solution is  $n = 1$ .

22. WLOG assume  $m \leq n$ . By the lifting the exponent lemma, we have

$$v_3(4^{3^m} - 1) = v_3(4 - 1) + v_3(3^m) = 1 + m.$$

Therefore,  $n + 2 \leq m + 1$ . This yields  $n < m$ , which is a contradiction. Thus, there is no solution.

23. Let  $p$  be the smallest prime divisor of  $n$ . Clearly,  $p$  is odd. Then  $2^n + 1 \equiv 0 \pmod{p}$  so that  $2^{2n} \equiv 1 \pmod{p}$ . Let  $d$  be the order of 2 modulo  $p$ . Then we have  $d \mid 2n$  since  $2^{2n} \equiv 1 \pmod{p}$ , and  $d \mid p - 1$  by Fermat little theorem. Note that

$$(2n, p - 1) = (2, p - 1) = 2$$

as  $p$  is the smallest prime divisor of  $n$ . This implies  $d \mid 2$ . The case  $d = 1$  is rejected since  $2 \not\equiv 1 \pmod{p}$ . For  $d = 2$ , we have  $2^2 \equiv 1 \pmod{p}$  so that  $p = 3$ .

Let  $v_3(n) = a$ . By the lifting the exponent lemma, we have

$$v_3(2^n + 1) = v_3(8^{\frac{n}{3}} + 1) = v_3(8 + 1) + v_3\left(\frac{n}{3}\right) = a + 1.$$

As  $2^n + 1$  is divisible by  $n^2$ , we have  $3^{2a} \mid 3^{a+1}$ . This implies  $2a \leq a + 1$ . The only possibility is  $a = 1$ .

One can check that  $n = 3$  is a solution. If  $n > 3$ , we let  $q$  be the second smallest prime divisor of  $n$ . Then  $2^n + 1 \equiv 0 \pmod{q}$  so that  $2^{2n} \equiv 1 \pmod{q}$ . Let  $f$  be the order of 2 modulo  $q$ . Then  $f \mid 2n$  and  $f \mid q - 1$ . This time we have  $(2n, q - 1) \mid 6$ . In any case, we must have  $2^6 \equiv 1 \pmod{q}$ . The only possibility is  $q = 7$ . However, we check that  $7 \nmid 2^n + 1$  since  $2^n + 1 \equiv 2, 3, 5 \pmod{7}$ .

Therefore, the only solution is  $n = 3$ .



## Valuation

24. There are  $\left\lfloor \frac{72}{2^k} \right\rfloor$  numbers from 1 to 72 which are multiples of  $2^k$ . Therefore, we have

$$\begin{aligned} v_2(72!) &= \left\lfloor \frac{72}{2} \right\rfloor + \left\lfloor \frac{72}{2^2} \right\rfloor + \left\lfloor \frac{72}{2^3} \right\rfloor + \left\lfloor \frac{72}{2^4} \right\rfloor + \left\lfloor \frac{72}{2^5} \right\rfloor + \left\lfloor \frac{72}{2^6} \right\rfloor \\ &= 36 + 18 + 9 + 4 + 2 + 1 = 70. \end{aligned}$$

So the largest  $n$  is 70.

25. As  $5 \mid 1001 - 101$ , we have

$$v_5(1001^{200!} - 101^{200!}) = v_5(1001 - 101) + v_5(200!) = 2 + v_5(200!)$$

by the lifting the exponent lemma. Note that

$$v_5(200!) = \left\lfloor \frac{200}{5} \right\rfloor + \left\lfloor \frac{200}{5^2} \right\rfloor + \left\lfloor \frac{200}{5^3} \right\rfloor = 40 + 8 + 1 = 49.$$

Hence, the largest  $n$  is  $2 + 49 = 51$ .

26. Let  $t$  be the nonnegative integer for which  $2^t \leq k < 2^{t+1}$ . We claim that  $n = 2^{m+t} - 2^t + k$  is a possible choice. Firstly, it is clear that  $n \geq k$ . Next, we consider the number of factors 2 appearing in each term of

$$\binom{2^{m+t} - 2^t + k}{k} = \frac{(2^{m+t} - 2^t + k)(2^{m+t} - 2^t + k - 1) \cdots (2^{m+t} - 2^t + 1)}{k(k-1) \cdots 1}.$$

For each  $j = 1, 2, \dots, k$ , let  $2^c \parallel j$ . If  $j \neq 2^t$ , then  $c \leq t-1$ . Note that  $2^c \mid 2^{m+t} - 2^t + j$  as  $2^c \mid j$ . Also, if  $2^{c+1} \mid 2^{m+t} - 2^t + j$ , then  $2^{c+1} \mid j$ . This contradicts the maximality of  $c$ . Hence,  $2^c \parallel 2^{m+t} - 2^t + j$ . In other words,  $2^{m+t} - 2^t + j$  in the numerator has the same number of factors 2 as  $j$  in the denominator if  $j \neq 2^t$ .

For  $j = 2^t$ , we have the term  $2^{m+t} - 2^t + j = 2^{m+t}$  in the numerator and the term  $j = 2^t$  in the denominator. This shows the numerator has  $m$  more powers of 2 than the denominator. Thus,  $\binom{n}{k} = 2^m a$  for some odd number  $a$  for our choice of  $n$  as desired.

27. Let  $a = 3^m \cdot c$  and  $b = 3^n \cdot d$  where  $c, d$  are not divisible by 3. If  $(15a + b)(a + 15b)$  is a power of 3, then  $15a + b$  is a power of 3. This implies

$$3^{m+1} \cdot 5c + 3^n \cdot d = 3^x \tag{1}$$

for some positive integer  $x$ . Note that

$$v_3(3^{m+1} \cdot 5c) = m + 1, \quad v_3(3^n \cdot d) = n, \quad v_3(3^x) = x.$$

The smallest two of these numbers are equal. It is obvious from (1) that  $x > m + 1, n$ . Therefore, we must have  $m + 1 = n$ . Similarly, by considering  $a + 15b$ , we obtain  $n + 1 = m$ . This is a contradiction. Hence  $(15a + b)(a + 15b)$  cannot be a power of 3.

28. There is no solution when  $k = 1$ . When  $k = 2$ , we have  $2m^2 = n$ . So  $m$  can be any positive integer and  $n = 2m^2$ .

Suppose  $k \geq 3$ . Clearly,  $n$  is an even number. As 4 divides the right-hand side,  $m$  is even. Also, it is obvious that  $m$  and  $n$  have the same set of odd prime divisors.

Consider any odd prime divisor  $p$  of  $m$  and  $n$ . Let  $v_p(m) = a$  and  $v_p(n) = b$ . By comparing the exponents of  $p$  on both sides, we get  $ak = b(k-1)$ . As  $(k, k-1) = 1$ , we have  $k-1 \mid a$ . Let  $a = (k-1)c$ . Then we get  $b = kc$ .

For the prime divisor 2, we let  $v_2(m) = u$  and  $v_2(n) = v$ . Then we have  $uk+1 = v(k-1)$ . Note that  $1 \equiv -v \pmod{k}$ . Let  $v = wk - 1$ . Then

$$u = \frac{v(k-1) - 1}{k} = \frac{(wk-1)(k-1) - 1}{k} = wk - w - 1.$$

Combining these, we can write  $m = 2^{wk-w-1}t^{k-1}$  and  $n = 2^{wk-1}t^k$  for some positive integers  $w$  and  $t$ . We check that

$$2m^k = 2^{1+(wk-w-1)k}t^{(k-1)k} = 2^{(wk-1)(k-1)}t^{k(k-1)} = n^{k-1}.$$

So any such pair is a solution. Note that the solution for the case  $k = 2$  is basically the same as this.

Therefore, the solutions are  $(2^{wk-w-1}t^{k-1}, 2^{wk-1}t^k, k)$  for  $k \geq 2$ .

29. Note that  $a \mid b^3$  and  $b \mid a^3$ . This shows  $a$  and  $b$  have the same set of prime divisors. Consider any prime divisor  $p$  of  $a$  and  $b$ . Then  $p \nmid c$  since  $(a, b, c) = 1$ . Let  $v_p(a) = \alpha$  and  $v_p(b) = \beta$ . Then

$$v_p(abc^3) = \alpha + \beta, \quad v_p(a^3) = 3\alpha, \quad v_p(b^3) = 3\beta.$$

The smallest two among these terms are equal. Thus, either  $\alpha = 2\beta$  or  $2\alpha = \beta$ . We partition the prime divisors according to these two kinds. This shows there are relatively prime positive integers  $m$  and  $n$  such that  $a = m^2n$  and  $b = mn^2$ .

The equation becomes  $m^3n^3c^3 = a^3 + b^3$ , i.e.  $(mnc)^3 = a^3 + b^3$ . There is no positive integer solution by Fermat last theorem.